How much does the scale cut depend on its fiducial?

The UV-fraction cut computed on the linear-bias fiducial against the paper model's own, in kept band powers and figures of merit

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1 Summary

The question. The forecast removes small scales with the UV-fraction cut of Saraivanov et al. [1]: a band power is kept while less than five percent of it comes from modes above a redshift-scaled wavenumber threshold (Section 3 defines every ingredient). Because the cut splits the *model's own* spectra, it must be computed on some model at some parameter point – a fiducial, the single point in parameter space at which a forecast is evaluated. The forecast computes the cut on the model it analyzes. The paper notebook computes it on the paper model's fiducial – the fiducial model: one-loop galaxy bias (linear bias plus the second-order bias terms, a counterterm and stochastic terms; the model is built in `notes/higher_order_bias`) with per-group offsets and tilts, the line-of-sight counterterm α_2 fixed at zero and the Fingers-of-God dispersion priors (on σ_v, the velocity spread that smears each bin's galaxies along the line of sight) widened by a factor of six. The linear-bias experiments keep a cut computed on the linear fiducial, where each bin carries only its linear galaxy bias b_1 (the coefficient relating the galaxy density contrast to the matter one on large scales). This note measures what that consistency is worth: how much the cut, and the science downstream of it, move between the two fiducials.

The answer.

1. **The two fiducials differ in exactly two numbers per bin.** At the paper model's fiducial point every parameter beyond linear bias is either zero (the counterterms, the stochastic terms, magnification) or shared with the linear fiducial (the cosmology, b_1, the dispersions, redshift-space distortions). What remains are the second-order biases b_2 and b_{s^2}, sitting on their calibration relations (Section 4).

2. **The cut moves by about one band per affected spectrum.** The kept band powers change by 1–3% at $k_{\max} \leq 0.1\,h/\mathrm{Mpc}$ on every layout (Figure 1). Auto spectra only lose bands, almost always one; cross-bin pairs move both ways, and a few pairs whose spectra change sign swing by tens of bands at $k_{\max} \geq 0.175\,h/\mathrm{Mpc}$ under either fiducial.

3. **The science moves less.** The paper model's FoM(w_0, w_a) (the dark-energy figure of merit; the setup paragraph below defines it) under the paper-fiducial cut is 0.95–1.00 of its value under the linear-bias cut on the main layouts, with one outlier of 0.85 on the coarsest layout at its smallest useful cut (Figure 2). The statistic η, which measures how well the quoted error bars cover where the fit lands, changes by at most 0.004, the

coverage by at most 0.003, the largest parameter shift by 0.034σ, and no scale cut changes between passing and failing the bias tests.

4. **One parameter carries the change.** Switching on b_2 alone reproduces 96–100% of the moved bands on $\Delta z = 0.05$ and 67–77% on $\Delta d_{\rm com} = 62$ Mpc; b_{s^2} alone moves at most four bands (Section 6).

5. **Within its own posterior, b_2 moves the cut ten to thirty times more.** Offsets of b_2 at ± 2 prior widths – excursions the data barely resist at the cuts that matter – change the kept band powers by 19–35%, against 1–3% between the two fiducials (Section 7).

The consistent choice is adopted. The paper notebook, `EXAMPLE_EVALUATE_FISHER_MODEL.ipynb`, computes its cut on the paper model's fiducial at the final derivative tier (the accuracy settings every published derivative uses; Section 2) and caches it (`chains/model_scale_cuts_final.npz`), and every Fisher result of the paper model uses that cache; the linear-bias notebooks keep the cut computed on the linear fiducial for the linear experiments. The numbers above are the price: a few percent of either figure of merit on the main layouts, 15% on the coarse control at its smallest useful cut, and no pass/fail change. What the strong b_2 dependence within the posterior implies stays open (appendix "Open issues").

The setup. The data are the DESI angular clustering band powers C_ℓ^{gg} alone, computed without the Limber approximation (which ties each multipole to a single wavenumber and fails for thin bins) [2], on four bin layouts: $\Delta d_{\rm com} = 62$ Mpc (equal comoving width) and $\Delta z = 0.02$, 0.05 and 0.2 (fixed redshift width). The first three are the main layouts; $\Delta z = 0.2$ is the coarse control. Here w_0 and w_a parameterize the dark-energy equation of state $w(a) = w_0 + w_a(1-a)$; Ω_m is the matter density today in units of the critical density; $A_{\rm s}$ is the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_{\rm s}$; and $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized (w_0, w_a) ellipse, likewise $\mathrm{FoM}(\Omega_m, 10^9 A_{\rm s})$. The paper model is scored on the pessimistic linear-bias prior throughout: each group's linear biases may wiggle by 3% per bin around a free line in redshift, with a short correlation length (the prior is derived in the correlated-prior note, `notes/correlated_bias_prior_v2`). Every number of this note is a property of one comparison: the same model, the same derivative caches (a Fisher forecast needs only the derivatives of the data vector at the fiducial, computed once and stored; `notes/fisher_machinery` builds that machinery) and the same covariance, under two sets of kept rows. The per-cut values behind both figures are in the figure folders' `data.json`.

2 Fiducial model and parameter values

This note compares two fiducials and the two scale-cut caches computed on them. This section freezes both, so the record does not depend on configuration files that can change.

Cosmology (shared by both fiducials). The evaluate override of the example yamls: $10^9 A_{\rm s} = 2.1$, $n_s = 0.96605$, $H_0 = 67.32\,\mathrm{km/s/Mpc}$, $\Omega_b = 0.04$, $\Omega_m = 0.3$, $w_0 = -1$, $w_a = 0$, $\sum m_\nu = 0.06\,\mathrm{eV}$ and $\tau = 0.0697186$ fixed; the derived $\sigma_8 = 0.8278$.

The linear fiducial. Growth-tied linear bias, $b_{1,i} = B_g/D(\bar{z}_i)$ per bin, with the group amplitudes $B_g = 1.34$ (BGS), 1.70 (LRG1–LRG3) and 0.84 (ELG1, ELG2) and D read off the linear power spectrum at $k = 0.005\,\mathrm{Mpc}^{-1}$; no higher-order term. The Fingers-of-God dispersions sit at 371, 358, 357, 348, 343, 304 and 289 km/s for the seven dispersion groups (BGS split at $\bar{z} = 0.255$). This is the override of `exploration/EXAMPLE_EVALUATE_LINEAR1–4.yaml`.

The paper fiducial. The same cosmology, b_1 and dispersions, with the second-order biases on their calibrations: b_2 on the Lazeyras et al. fit $b_2(b_1)$ [4] and b_{s^2} on co-evolution, $b_{s^2} = -\frac{4}{7}(b_1 - 1)$ [6]; every offset, tilt, counterterm and stochastic amplitude is zero. This is the override of `EXAMPLE_EVALUATE1–4.yaml` (Section 4).

The two cut caches. Both caches hold the UV-fraction cut with $\alpha_{\mathrm{UV}} = 0.05$ at $k_{\max} = 0.05$ to $0.25\,h/\mathrm{Mpc}$ (results use $k_{\max} \leq 0.2$), intersected with the shipped mask (the fixed per-bin mask distributed with each layout's dataset; Section 3):

- **the paper-fiducial cache**, computed on the paper fiducial at this study's reference settings (CosmoLike, the compiled code that projects the power spectrum onto the sky, at accuracy boost 4 with integration accuracy 2, the non-Limber χ grid at four times the sampling examples' factors, the low-rank engine – the project's non-Limber method – at rank 8 with multipole cap 2000, and CAMB, the code that computes the matter power spectrum, at AccuracyBoost 2 with `k_per_logint` 30 and transfer k up to $80\,\mathrm{Mpc}^{-1}$). The paper notebook's own cache, `chains/model_scale_cuts_final.npz`, holds the same cut at the final derivative tier; the two differ by one kept band, on $\Delta z = 0.2$ at $k_{\max} = 0.075\,h/\mathrm{Mpc}$;
- **the linear cache**, `chains/linear_scale_cuts_final.npz`, computed on the linear fiducial at the final derivative tier (accuracy boost 3 with integration accuracy 1, `nonlimber_accuracyboost` 64, CAMB AccuracyBoost 2 with `k_per_logint` 30), with the paper data sets swapped in.

Both fiducials were also run at the study's reference settings for the like-for-like comparison of Section 4.

Scoring. Every figure of merit is the paper model (the fiducial model: per-group offsets and tilts, with $\alpha_2 = 0$ in every bin and the seven σ_v prior widths six times the exploration ones) on the pessimistic linear-bias prior, scored from the paper model's derivative base at the final tier (five-point stencils at relative step $h = 0.02$, `nonlimber_accuracyboost` 64, patched halofit tolerance) and the Gaussian covariance of the paper data sets, built on the one-loop fiducial with massless neutrinos over 14,000 deg^2, its inverse rescaled by 12,300/14,000. The final tier is the accuracy bundle that the convergence study of `notes/fisher_machinery` fixed for every published derivative; a five-point stencil is the five-evaluation finite difference it uses in place of a simple two-sided one.

3 The UV-fraction scale cut

Band powers. The data vector holds angular power spectra C_ℓ^{gg}: the clustering of each redshift bin with itself (the autos) and with its listed neighbours (the cross pairs), as functions of the multipole ℓ, the angular wavenumber of a spherical-harmonic mode. The spectra are averaged over bands of width $\Delta\ell = 10$; one such average is a *band power*, the unit in which bands are kept or cut.

The IR/UV split. Each C_ℓ is an integral over wavenumbers k and redshifts of the three-dimensional galaxy power spectrum $P_{gg}(k, z)$. The cut splits that integral in two with a step function on k:

$$C_\ell \;=\; C_\ell^{\mathrm{IR}} \;+\; C_\ell^{\mathrm{UV}}, \tag{1}$$

where C_ℓ^{IR} is computed with P_{gg} multiplied by a step that keeps $k \le k_{\max}/D(z)$, and C_ℓ^{UV} is the rest. The names are physics shorthand: infrared (IR) for large scales, small k; ultraviolet (UV) for small scales, large k, where the model is least trusted. $D(z)$ is the linear growth factor, normalized to $D(0) = 1$, so the threshold $k_{\max}$ is quoted at $z = 0$ and grows with redshift as $1/D(z)$, following the nonlinear scale. In practice each bin i is cut at one threshold, $k_{\max}/D(\bar{z}_i)$ with $\bar{z}_i$ its mean redshift, and a cross pair at the midpoint of its two bins.

The criterion. A band power's *UV fraction* is the share of its signal that comes from above the threshold,

$$\mathrm{uvfrac}(\ell) \;=\; \frac{|C_\ell^{\mathrm{UV}}|}{|C_\ell^{\mathrm{UV}}| + |C_\ell^{\mathrm{IR}}|}, \tag{2}$$

with absolute values so that a negative term counts like a positive one. A spectrum's kept bands run from its first band up to, and not including, the first band where uvfrac $\ge \alpha_{\mathrm{UV}}$; every later band is dropped even if its fraction dips back below (the prefix rule: the kept bands are one unbroken run from the first band). The baseline threshold is $\alpha_{\mathrm{UV}} = 0.05$ [1], following DeRose and Chen [5]. Throughout, "kept" means a band the cut keeps *and* the shipped mask keeps – the fixed per-bin mask distributed with each layout's dataset, a cut at $k = 0.1185\,\mathrm{Mpc}^{-1}$ ($0.176\,h/\mathrm{Mpc}$) in every bin – because only those rows ever enter a Fisher matrix.

Where it runs. The split is implemented as a k window inside the compiled spectra themselves (`cosmolike/cosmo2D_scuts.c`): the step multiplies P_{gg} inside every galaxy-clustering engine – the Limber approximation, the non-Limber paths [2] and the cross-pair engine. The cut therefore sees every term the compiled model puts into C_ℓ^{gg}, with the modes a thin bin's radial structure adds counted toward the UV part. Because Eq. (1) splits the model's own spectra, the cut depends on the model and the parameter point it is evaluated at: change the fiducial and the kept bands can change.

A two-term toy. How a second term moves the cut is pure arithmetic. Take one spectrum, two of its bands and two terms of the model: a linear-bias term (b_1^2 times the matter power) and a small one-loop term (the b_2-type corrections), in units where the linear term totals 100. The loop term is worth 2 on both bands – but the two terms have different reach in k. The

	band A (low ℓ)	band B (near the cut)
linear term, IR / UV	99.0 / 1.0	95.2 / 4.8
loop term, IR / UV	1.9 / 0.1	1.0 / 1.0
UV fraction, linear alone	0.010	0.048 (kept)
UV fraction, with the loop	0.011	0.057 (cut)

Table 1: A worked toy of the UV-fraction criterion: two bands of one spectrum, a linear-bias term and a b_2-like loop term worth 2% of it, with the threshold $\alpha_{\rm UV} = 0.05$.

term of C_ℓ^{gg}	inside the cut?	at the paper fiducial
$b_1^2\times$ matter power	yes	identical to the linear fiducial
one-loop b_2, b_{s^2} terms	yes	on their calibrations (*the difference*)
their high-k constant P_∞	yes	set by $(b_2 + \frac{2}{3}b_{s^2})^2$
counterterm $\alpha_0 k^2 P$	yes	0
counterterm α_2	yes, when on	0 and fixed
redshift-space distortions	yes	identical dispersions
magnification	yes	0
stochastic templates SN0, SN2	**no** (added in Python)	0

Table 2: What the UV-fraction cut sees of the one-loop model, and what each term is worth at the paper model's fiducial point.

linear term is IR-dominated everywhere; the loop term grows with k, so at the band near the threshold half of it sits above the cut. Table 1 does the bookkeeping of Eqs. (1)–(2).

At band A nothing happens: the fraction moves from 0.010 to $1.1/102 = 0.011$. At band B the same 2% term flips the band: the fraction moves from 0.048, just under the threshold, to $5.8/102 = 0.057$, and band B – and with it, by the prefix rule, every band after it – is cut. Two lessons carry over to the real spectra. First, the criterion responds to a term's *reach in* k, not its size: a 2% term whose UV share is ten times the linear term's moves the fraction by nearly a fifth. Second, only bands already close to the threshold flip, which is why the real changes below are one band per affected spectrum, not wholesale. The sign does not matter: by the absolute values in Eq. (2), a negative UV-heavy term (such as the constant P_∞ below) raises the fraction just as a positive one does.

4 What the cut sees

Every compiled term. Because the k window acts inside the compiled engines, the cut sees the full one-loop model: the linear term b_1^2 times the nonlinear matter power; the one-loop bias terms in b_2 (the second-order density bias) and b_{s^2} (the tidal bias), including the constant they leave at high k, $P_\infty \propto -(b_2 + \frac{2}{3}b_{s^2})^2$, a negative shot-noise-like term; the density counterterm $\alpha_0 k^2 P$; the line-of-sight counterterm α_2 [3] when it is switched on; linear redshift-space distortions on each bin's Fingers-of-God–blurred redshift distribution; and magnification (the change of a bin's observed counts by gravitational lensing of foreground matter, with amplitude zero at both fiducials). Table 2 lists each term with its value at the paper fiducial.

What it cannot see. The two stochastic templates – SN0, a white-noise (constant in k) power, and SN2, its $k_\parallel^2$ companion – are added on the Python side of the likelihood, after the

layout	$k_{\max} = 0.075\,h/\mathrm{Mpc}$	0.1	0.2
$\Delta d_{\mathrm{com}} = 62$ Mpc	-1.0%	-1.4%	-0.1%
$\Delta z = 0.02$	-1.1%	-0.6%	-0.2%
$\Delta z = 0.05$	-3.2%	-1.3%	-2.0%
$\Delta z = 0.2$	-1.4%	$+0.0\%$	$+5.5\%$

Table 3: Change of the kept band powers (cut intersected with the shipped mask), linear-bias cut to paper-fiducial cut. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

compiled vector. The cut never evaluates that code, so it is blind to them. At the fiducial both are zero, so nothing is lost here; but a fiducial with nonzero stochastic power would need care, because a constant 3D power in a thin bin is dominated by radial modes and carries a large UV fraction the cut would not count.

The two fiducials. It follows that at the fiducial point the paper model differs from the linear-bias model in exactly one respect: b_2 and b_{s^2} sit on their *calibration relations* – fixed functions of b_1 measured in simulations, used as the center the sampled offsets move around. Here b_2 follows the Lazeyras et al. fit $b_2(b_1)$ [4] and b_{s^2} the coevolution relation $b_{s^2} = -\frac{4}{7}(b_1 - 1)$ [6]; on $\Delta d_{\mathrm{com}} = 62$ Mpc the bin values run from $b_2 = -0.80$ (ELG1) to $+2.04$ (LRG3), and b_{s^2} from -1.06 to -0.15. Everything else – cosmology, redshift distributions, b_1, dispersions, settings – is identical. Numerical settings contribute nothing to the comparison: recomputed at the study's reference settings, the linear-bias cut equals the cached cut of the linear-bias Fisher results on every kept row of every layout, and the cut alone differs by one band on two of twenty-eight layout–cut combinations.

5 Results

The cut moves by about one band. Figure 1 shows the change of the kept band powers when the cut's fiducial becomes the paper model's; Table 3 gives the totals at three cuts. The autos only lose bands, almost always exactly one ($\Delta\ell = 10$) and never more than three; the movers are the bins where the loop terms are largest near the threshold – LRG3, with b_2 up to $+2$, and the ELG bins, with $b_2 \approx -0.8$ and the largest $|P_\infty|$. The mechanism is the toy's: at the last kept band the one-loop terms change the full C_ℓ by -4 to $+6\%$ at $k_{\max} = 0.075\,h/\mathrm{Mpc}$, either sign adds scale dependence that grows with k, and the UV fraction of the bands the linear cut keeps rises by a median factor of 1.00–1.05. A band sitting just under α_{UV} flips; the rest stay.

Fragile cross pairs. The cross pairs move both ways, by one band at $k_{\max} \leq 0.1\,h/\mathrm{Mpc}$. At 0.175 and above on the wider layouts, a handful of adjacent pairs swing by tens of bands: these are spectra that change sign along ℓ, so $|C_\ell|$ passes near zero, the UV fraction crosses α_{UV} at a low band, and the prefix rule then removes everything after it. A percent-level change of the model moves that crossing by many bands in either direction – the $+5.5\%$ of $\Delta z = 0.2$ at $0.2\,h/\mathrm{Mpc}$ is one such pair keeping 36 bands instead of 3. The linear-bias cut of these pairs is exactly as fragile as the paper-fiducial one; neither is better defined.

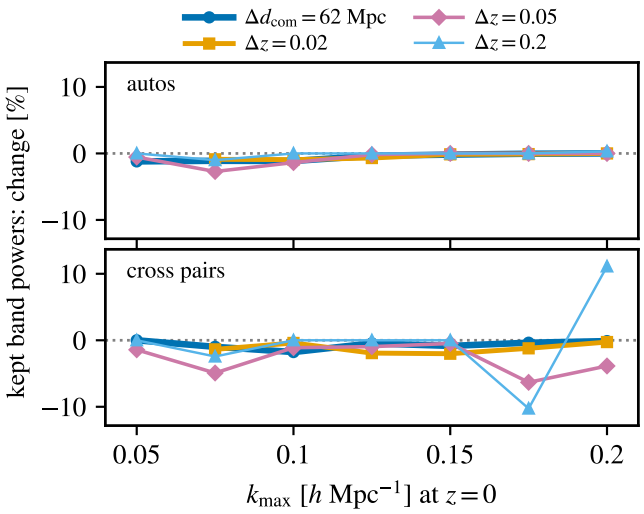


Figure 1: Change of the band powers kept by the UV-fraction scale cut when its fiducial becomes the paper model's instead of the linear-bias one, for the clustering autos (top) and the cross-bin pairs (bottom), per layout (color, line width and marker; legend). The dotted line marks no change. Every count intersects the cut with the shipped mask; $\Delta z = 0.02$ keeps no band at $0.05\,h$/Mpc. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

The science. Figure 2 shows what the moved bands are worth. On the main layouts the paper model's figures of merit under the paper-fiducial cut are 0.95–1.00 of their values under the linear-bias cut: 1–3% lower on the thin layouts at every cut, with the largest single cost 5.3% in FoM(w_0, w_a) and 6.4% in FoM$(\Omega_m, 10^9 A_{\rm s})$ on $\Delta z = 0.05$ at $0.1\,h$/Mpc. The honesty of the fit does not move: $\eta = \sqrt{\langle \chi^2 \rangle / n_{\rm dof}}$, the check that the quoted error bars cover where the fit lands under the known drifts plus the prior scatter, stays between 1.000 and 1.031 under both cuts and changes by at most 0.004; the coverage of the 68.3% region changes by at most 0.003. The largest change of any tested parameter shift is 0.034σ. The break points of the bias tests – the largest cut at which every tested truth (the paper notebook's fixed drifts and offsets, data the model does not contain; `notes/higher_order_bias` lists them) shifts the fit by less than $\sigma/3$ – are identical under both cuts on every layout: no cut moves from pass to fail or back.

The one large move. The outlier is $\Delta z = 0.2$ at $0.075\,h$/Mpc: the FoM ratio is 0.85. Four bands go there – one each from a BGS and an LRG3 auto, one from each of two LRG3×ELG1 pairs – and with ten wide bins and few bands per bin at that cut, those four carry a large share of the information ($\sigma(\Omega_m)$ moves from 0.581 to 0.671). The +4% of the same layout at $0.2\,h$/Mpc is the fragile sign-changing pair above, not the bias model.

The adopted choice. The forecast computes the cut on the model it analyzes. The paper notebook computes its cut on the paper model's fiducial and caches it (Section 2); every Fisher result of the paper model uses that cache. The linear-bias notebooks keep the cut computed on the linear fiducial for the linear experiments. The numbers of this section are the price of

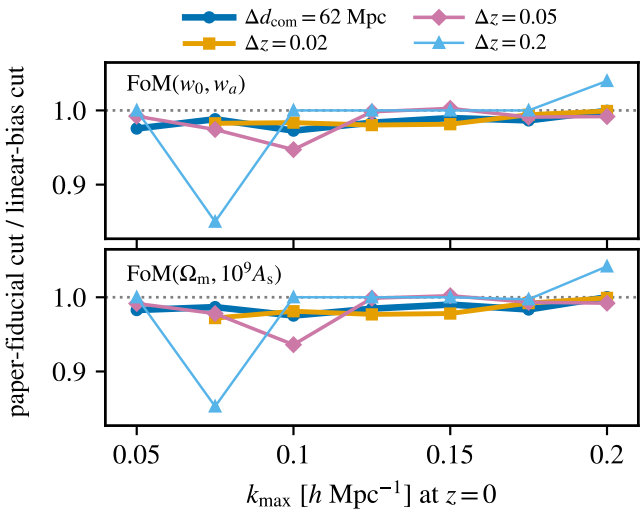


Figure 2: The paper model's figures of merit under the paper-fiducial cut over those under the linear-bias cut, FoM(w_0, w_a) (top) and FoM($\Omega_m, 10^9 A_{\rm s}$) (bottom), per layout (color, line width and marker; legend). The dotted line marks a ratio of one. $\Delta z = 0.02$ has no fit at $0.05\,h$/Mpc; each cut is intersected with the shipped mask. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

	$\Delta d_{\rm com} = 62$ Mpc		$\Delta z = 0.05$	
kept bands moved by	$0.075\,h$/Mpc	0.1	0.075	0.1
b_2 alone	−12	−30	−26	−16
b_{s^2} alone	−1	−4	−1	−1
both (paper fiducial)	−18	−39	−27	−16

Table 4: Kept band powers moved against the linear-bias cut when each second-order bias is switched on alone, and when both are (the paper fiducial). $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

that consistency: a few percent of either figure of merit on the main layouts and 15% on the coarse control at its smallest useful cut, with η, the coverage and every pass/fail conclusion unchanged. The cut cache is tied to its fiducial: when the fiducial model changes, the cut is recomputed and the cache replaced.

6 Attribution: b_2 carries the change

One term at a time. Since the two fiducials differ only through b_2 and b_{s^2}, the change can be attributed by switching the two on one at a time: once with b_2 on its calibration and $b_{s^2} = 0$, once the reverse. Table 4 shows the moved bands against the linear-bias cut. b_2 alone reproduces 96–100% of the change on $\Delta z = 0.05$ at 0.075–0.1 h/Mpc and 67–77% on $\Delta d_{\rm com} = 62$ Mpc; b_{s^2} alone moves at most four bands anywhere. The remainder is non-additive and comes from the b_2–b_{s^2} cross terms – the $b_2 b_{s^2}$ loop and the cross part of $P_\infty \propto (b_2 + \frac{2}{3} b_{s^2})^2$.

7 Robustness: the cut within the bias priors

Three shifted fiducials. The paper fiducial is one point; the analysis samples around it. Three coherent shifts of the paper fiducial, each at the scale its own prior allows, measure how stable the cut is inside that volume (all numbers at 0.075–0.1 h/Mpc on $\Delta d_{\rm com} = 62$ Mpc and $\Delta z = 0.05$, kept band powers against the paper-fiducial cut):

1. **Linear bias up** 6% (+2 widths of the pessimistic prior's per-bin residual, with b_2 and b_{s^2} following their calibrations): the cut keeps +1.0 to +2.1% more.
2. **The b_2 and b_{s^2} offsets at ± 2 prior widths** ($\Delta_2 = \pm 4$ and $\Delta_s = \pm 2$ in every group, the fiducial model's sampling freedom around the calibrations): the cut keeps 19–35% *fewer* band powers, under either sign. A b_2 four units off its calibration makes the loop terms a large share of P_{gg} near the threshold, so the UV fraction rises everywhere.
3. **The counterterm α_0 at +1 prior width** ($+16.1\,h^{-2}\,\mathrm{Mpc}^2$): the cut keeps 1.9–3.7% fewer. A k^2P term is pure UV-weighted power; this one shift, inside one prior width, moves the cut as much as the whole linear-to-paper change does.

Against what the data constrain. These are not academic excursions. At the cuts where the bias tests pass or nearly pass, the data barely measure the offsets: the paper model's marginalized errors on Δ_2 are 1.4–2.0 at 0.075 h/Mpc and 0.9–1.9 at 0.1, close to the prior width 2. An offset of 4 is therefore a plausible 2σ posterior excursion, and within it the cut moves ten to thirty times more than between the linear-bias and paper fiducials. Both fiducials are points inside that band.

Caveats. The shifts are rigid: every group moves together, by a prior-width multiple, which is rarer than a posterior draw of the same size; and nothing is refitted, so the quoted band-power changes measure the cut's sensitivity, not a constraint. The conclusion they support is deliberately limited: the choice between the two fiducials is a small effect *compared with* the cut's dependence on where the bias parameters actually sit.

A Method and checks

The cuts. Each cut is the project's own scale-cut call (`gg_scale_cuts_tomo`, the binding of `cosmolike/cosmo2D_scuts.c`) run at the scale-cut study's reference settings, once per fiducial; only the parameter point changes between runs. Each run is validated in two ways: its unwindowed spectra equal the evaluated data vector bitwise on every shipped entry, and the linear-bias run equals the cached cut of the linear-bias Fisher results on every kept row of every layout – settings noise flips at most one band of the cut alone, and never a kept row.

The scoring. The paper model is rescored with the Fisher notebook's own code: the paper model's derivative base at the final tier (five-point stencils) and the same covariance, with only the kept rows exchanged. A different cut selects rows of the cached derivative matrices; nothing is recomputed, and every row a new mask keeps is checked to carry a nonzero fiducial entry and derivative row. On the paper-fiducial masks the scoring reproduces the paper notebook's figures of merit (`notes/bias_model_summary`) on every layout and cut except $\Delta z = 0.2$ at

$k_{\max} = 0.075\,h/\mathrm{Mpc}$, where the notebook's final-tier cut keeps one band that this study's reference-settings masks cut. η, the coverage and the break rule are the builder's own (known drifts plus the pessimistic prior scatter; a cut passes while every tested truth shifts the fit by less than $\sigma/3$).

B Open issues

- **What the b_2 dependence implies.** Within the b_2 offsets' own posterior the cut moves ten to thirty times more than between the two fiducials: the cut is a convention anchored at a parameter point, not a quantity the data pin down. Three responses are possible, each with a cost. Treat the fiducial cut as a fixed convention, like any other fiducial choice – simplest, but the kept data vector would have looked substantially different had the fiducial b_2 been drawn elsewhere in its posterior. Verify the headline results under cuts computed at shifted b_2 – bounded extra work, and this note's machinery does it, but it multiplies the cut caches. Or fold the cut's variation into a systematic budget – the most conservative, at the price of wider errors. The shifts tested here are rigid, coherent prior-width excursions; a posterior-draw version of the same test would sharpen the factor of ten to thirty before any of these responses is chosen.

References

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